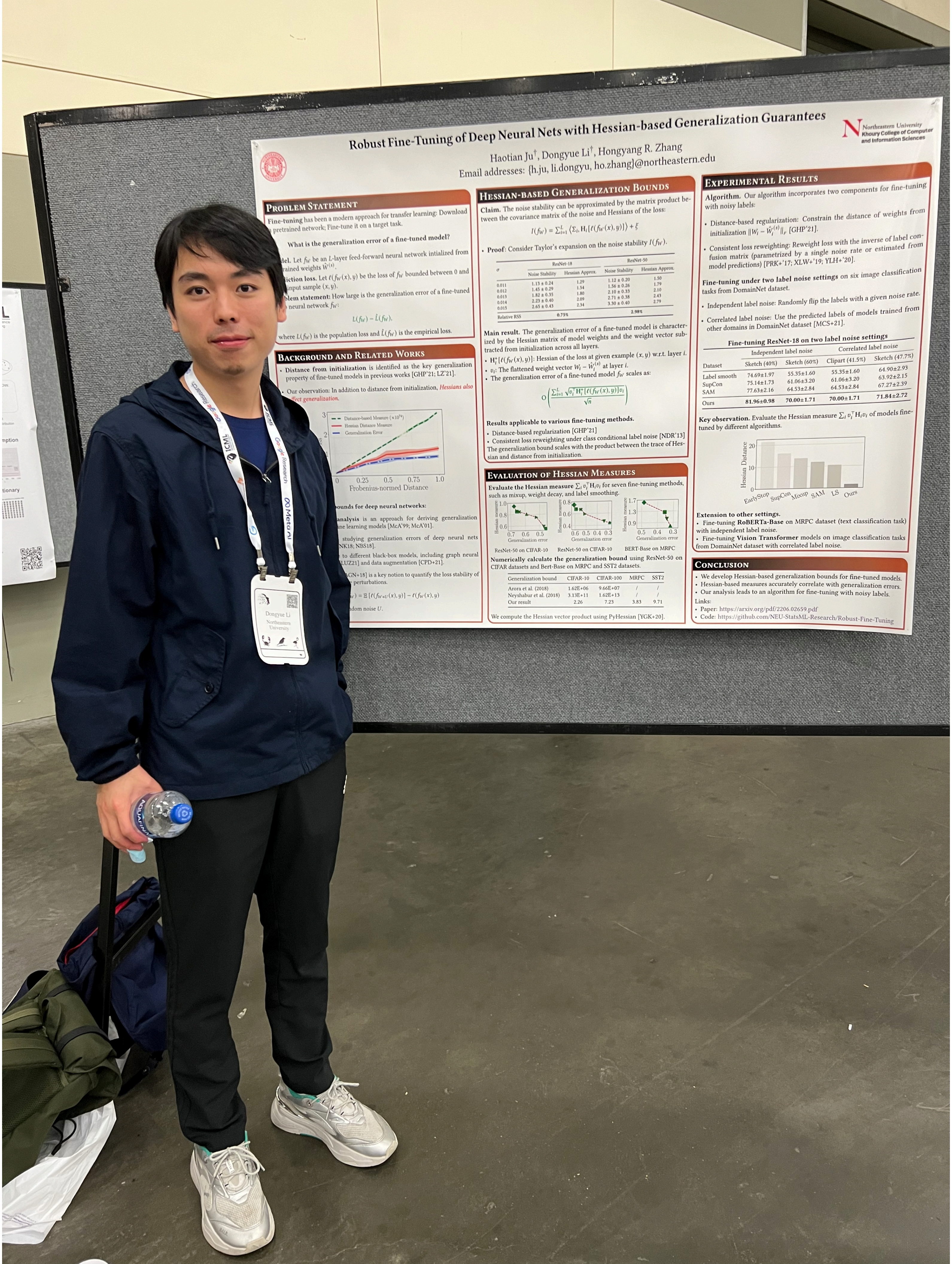
Robust Fine-Tuning of Deep Neural Nets with Hessian-based Generalization Guarantees
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Problem Statement
What is the generalization error of a fine-tuned model?
where L(fW) is the population loss and L̂(fW) is the empirical loss.
Background and Related Works
Distance from initialization is identified as the key generalization property of fine-tuned models in previous works [GHP'21; LZ'21].
Frobenius-normed Distance
Hessian-based Generalization Bounds
Claim. The noise stability can be approximated by the matrix product between the covariance matrix of the noise and Hessians of the loss:
Proof: Consider Taylor's expansion on the noise stability I(fW).
Main result. The generalization error of a fine-tuned model is characterized by the Hessian matrix of model weights and the weight vector subtracted from initialization across all layers.
The generalization error of a fine-tuned model fW scales as:
Results applicable to various fine-tuning methods.
Distance-based regularization [GHP'21]
Consistent loss reweighting under class conditional label noise [NDR'13]
Evaluation of Hessian Measures
ResNet-50 on CIFAR-10
ResNet-50 on CIFAR-10
BERT-Base on MRPC
Numerically calculate the generalization bound using ResNet-50 on CIFAR datasets and Bert-Base on MRPC and SST2 datasets.
We compute the Hessian vector product using PyHessian [YGK+20].
Experimental Results
Algorithm. Our algorithm incorporates two components for fine-tuning with noisy labels:
Distance-based regularization: Constrain the distance of weights from initialization [GHP'21].
Consistent loss reweighting: Reweight loss with the inverse of label confusion matrix (parametrized by a single noise rate or estimated from model predictions) [PRK+'17; XLW+'19; YLH+'20].
Fine-tuning under two label noise settings on six image classification tasks from DomainNet dataset.
Independent label noise: Randomly flip the labels with a given noise rate.
Correlated label noise: Use the predicted labels of models trained from other domains in DomainNet dataset [MCS+21].
Fine-tuning ResNet-18 on two label noise settings
Key observation. Evaluate the Hessian measure of models fine-tuned by different algorithms.
Hessian Distance
EarlyStop SupCon Mixup SAM LS Ours
Extension to other settings.
Fine-tuning RoBERTa-Base on MRPC dataset (text classification task) with independent label noise.
Fine-tuning Vision Transformer models on image classification tasks from DomainNet dataset with correlated label noise.
Conclusion
We develop Hessian-based generalization bounds for fine-tuned models.
Hessian-based measures accurately correlate with generalization errors.
Our analysis leads to an algorithm for fine-tuning with noisy labels.
Paper: https://arxiv.org/pdf/2206.02659.pdf
Code: https://github.com/NEU-StatsML-Research/Robust-Fine-Tuning
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